

Synthetic Handwriting with Generative Adversarial Networks

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Math 418

This project investigates conditional GANs (cGANs) for generating handwritten characters using the EMNIST Balanced dataset.

Two architectures are compared:

- Fully connected GAN (baseline)
- Convolutional GAN (improved spatial learning)

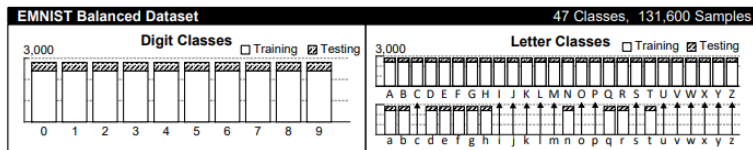
A classifier-based evaluation is used to measure synthetic image quality.

Introduction

Goal: generate realistic handwritten characters using GANs.

GAN structure:

- Generator: creates fake images
- Discriminator: distinguishes real vs fake



Dataset: EMNIST Balanced (47 classes: digits + letters)

cGAN Architecture

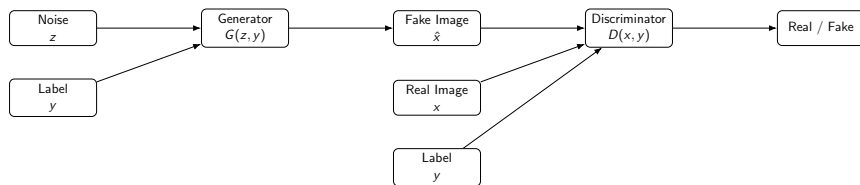


Figure: Conditional GAN (cGAN) architecture

Traditional GAN (No Convolution)

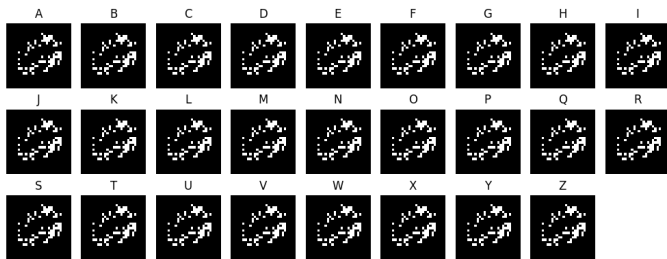
Hypothesis: Fully connected GAN will struggle with image structure.

Training setup:

- 200 epochs
- Batch size: 128
- Noise vector: 100D
- Adam optimizer (2×10^{-4})

Expected issue: No spatial feature learning

Baseline Results



Poor quality: outputs collapse and do not resemble characters

Convolutional GAN

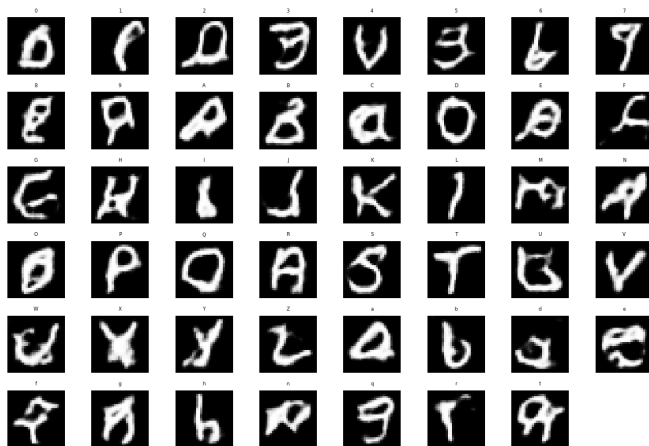
Why convolutional layers?

- Capture edges, strokes, curves
- Preserve spatial structure
- Better for image generation

Architecture:

- Generator: ConvTranspose layers
- Discriminator: CNN classifier

Conv GAN Results (Early)



After 5 epochs: already forming recognizable shapes

Conv GAN Results (Final)



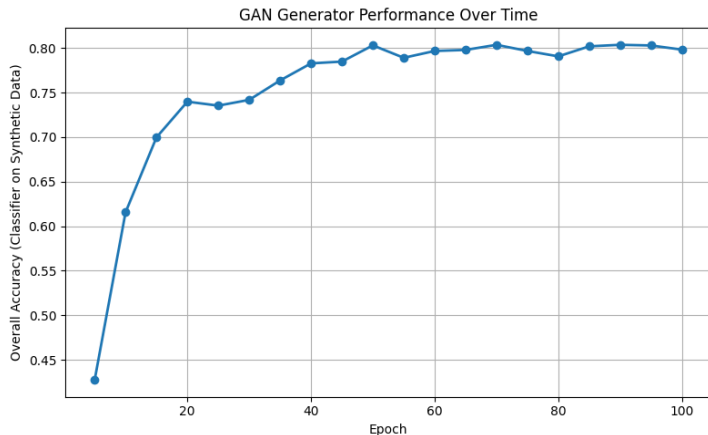
After 100 epochs: realistic handwritten characters

Image quality is subjective → use classifier-based scoring.

Pipeline:

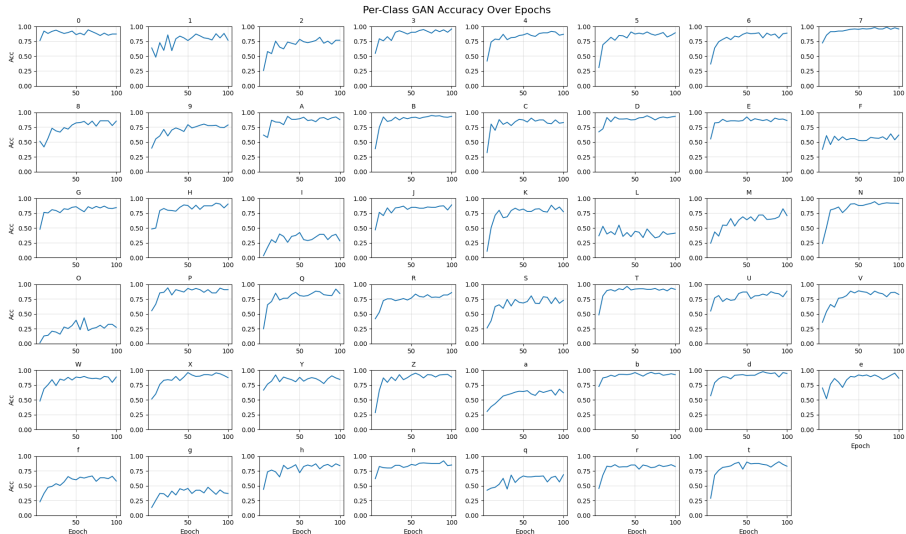
- Train classifier on real EMNIST
- Generate synthetic dataset
- Measure classification accuracy

Classification Accuracy



Accuracy increases over training → improved generator quality

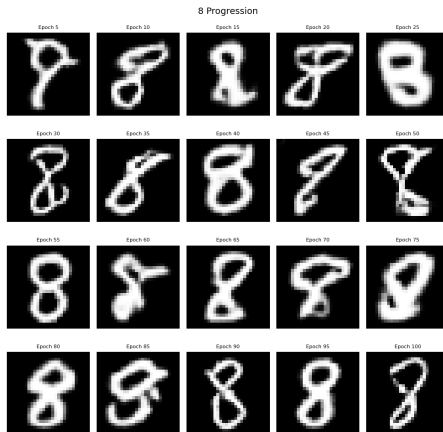
Per-Class Performance



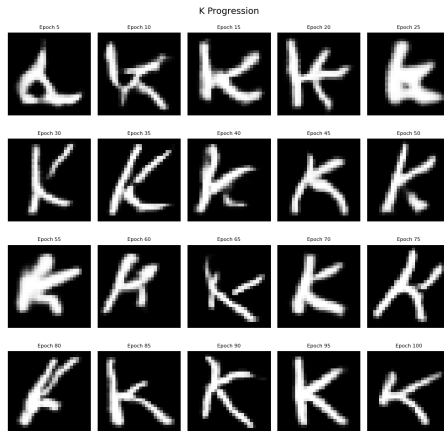
Confusions occur between similar characters (O vs 0, g vs 9, etc.)

Generation Progression

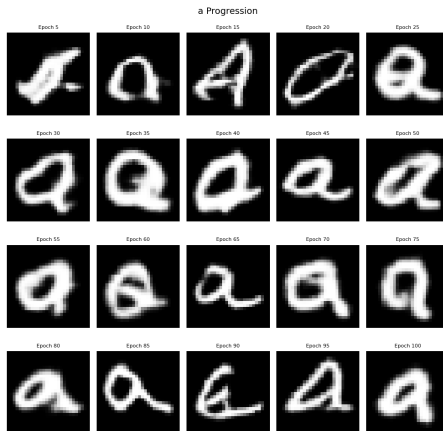
- Fixed noise vector across epochs



Generation Progression



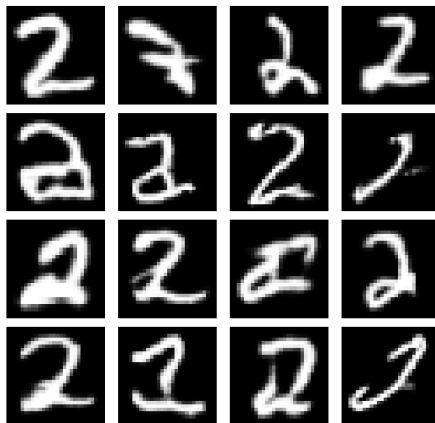
Generation Progression



Effect of Noise Variation

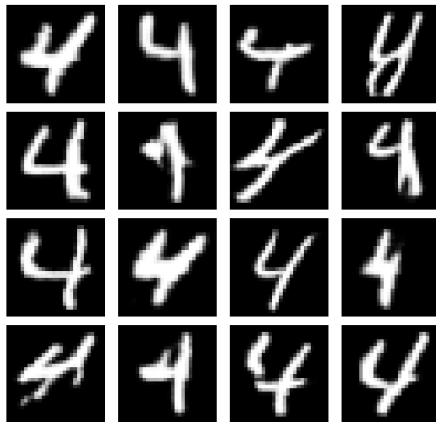
- Same class, different noise vectors

Generated Samples - Label 2



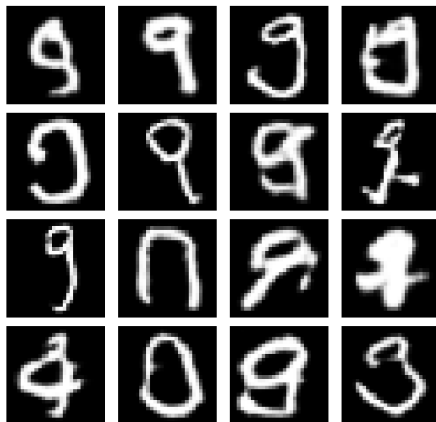
Effect of Noise Variation

Generated Samples - Label 4



Effect of Noise Variation

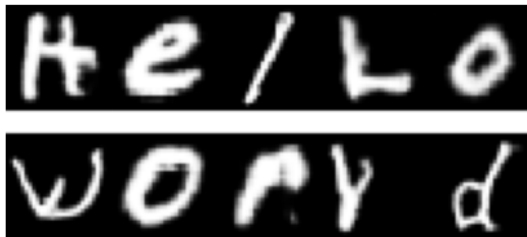
Generated Samples - Label g



Synthetic handwritten text generation:

- Words
- Digits
- Full sentences / quotes

Generated Examples



"Hello world"

Generated Quote

It is not
enough to
have a
good mind
the main
thing is
to use it
well/
REAL
Descartes

Further Study

- Personal handwriting style modeling
- Class-specific GANs
- Improved preprocessing
- Better class separation
- Noise-controlled style generation

Conclusion

- Convolutional GANs dramatically outperform fully connected GANs
- Generated samples become realistic over training
- Classifier provides useful quantitative evaluation
- EMNIST is a strong benchmark for generative models